2. Data Exploration

Dataset and respondent-level splitting

The released dataset contains 870 responses from 290 respondents, with three responses per respondent. The target is a three-class label: Baseball, Basketball, or Soccer. The 12 input features comprise one numerical ease-of-entry rating, three Likert questions, three multi-select questions, four free-text questions, and one ticket-price question entered as text.

Using random seed 42, we shuffled respondent IDs and assigned 203 respondents (609 rows) to training, 43 (129 rows) to validation, and 44 (132 rows) to an internal test set. Grouping by respondent prevents the same person's wording and answering habits from appearing across splits. ID is excluded from model inputs. All exploration and fitted preprocessing described below use training responses only; validation and internal test responses are reserved for later evaluation. The separate instructor-held test set is unavailable to us.

The training set contains exactly 203 responses per class (33.3% each), so class rebalancing is not currently required. A classifier that always predicts one class would obtain 33.3% training accuracy.

Feature types and missing values

We lowercased responses, normalized whitespace, and treated empty strings and explicit markers such as "n/a", "null", "-", and "[]" as missing. We retained "none" and "no particular type" because they can be meaningful answers. Missingness was low, ranging from 0.66% to 3.94%.

Feature	Type	Missing / 609	Missing (%)
Ease of entry	numeric	4	0.66
Strategy	ordinal	5	0.82
Injury risk	ordinal	4	0.66
Individual brilliance	ordinal	5	0.82
Playing location	multi-select	4	0.66
Season	multi-select	4	0.66
Music	multi-select	4	0.66
Food and drinks	text	10	1.64
Colour	text	5	0.82
Song	text	24	3.94
Viewing feelings	text	4	0.66
Ticket price	numeric text	4	0.66

There were no exact duplicate training rows and no repeated respondent-label pairs. Two answer-label combinations were identical across different respondents; we retained them because independently supplied identical answers can be legitimate. Ratings were checked against the 0-10 range and Likert responses against the five recognized options; neither check identified invalid nonmissing answers.

Observed distributions

Ease ratings covered 0-10, including fractional values, with class means of 4.13 for Baseball, 5.96 for Basketball, and 6.19 for Soccer. Agreement was common: "Agree" accounted for 49.0%, 38.0%, and 36.8% of nonmissing strategy, injury, and individual-brilliance answers, respectively. These features varied across respondents rather than being constant.

Multi-select responses contained 54 location combinations, 13 season combinations, and 96 music combinations. Text was diverse even after case and whitespace normalization: food, colour, song, and feeling had 435, 141, 348, and 329 distinct nonmissing answers. This supports decomposing option combinations and using a text representation rather than treating every complete response as an unrelated category.

Encoding, price handling, and scaling

We encoded Likert responses from 1 (Strongly disagree) to 5 (Strongly agree). This preserves ordering but assumes equal spacing; one-hot encoding remains an alternative for validation. Each multi-select option became a separate binary feature. Missing multi-select answers have zero option indicators plus a missingness flag, distinguishing them from observed selections.

For ticket prices, we extracted single amounts or stated caps and used the midpoint of simple ranges. Bare numbers and unspecified "$" amounts were assumed to be CAD. Explicit foreign currencies and ambiguous answers were left unparsed rather than converted or guessed. In total, 555/609 responses (91.1%) yielded prices. The remaining 54 comprise four explicitly missing and 50 unparseable responses. These rules are simplifying assumptions and the saved parsing audit supports manual review.

Parsed prices were right-skewed: class medians were CAD 50, 150, and 150 for Baseball, Basketball, and Soccer, while means were approximately CAD 86, 224, and 309. We flagged values above the training upper IQR fence of CAD 425 but retained them because expensive seats can be plausible. We applied log(1 + price) to reduce skew and the influence of large values.

Numerical and ordinal missing values were filled using training medians and accompanied by missingness indicators. We standardized these columns using training means and standard deviations after the price transformation, preventing larger numerical scales from dominating a distance-based or regularized linear model. Binary indicators and text vectors were not standardized.

Each of the four free-text fields was separately represented using TF-IDF unigrams, with at most 100 terms per field and a minimum document frequency of two training responses. TF-IDF multiplies term frequency by smoothed inverse document frequency, log((1 + n)/(1 + df)) + 1, and L2-normalizes each response vector. It retains word-level evidence across varied answers while reducing the weight of ubiquitous terms. Missing text becomes an empty string with a missingness flag. Vocabulary and IDF are fitted on training only; unseen words are ignored. Preprocessing must be refitted within each training fold if cross-validation is used.

Relationships with the label and implications

Season shows a clear class association (Figure 1). Winter was selected in 57.1% of Basketball responses, versus 6.9% for Baseball and 8.4% for Soccer. Summer was selected in about 92% of every class, so its binary indicator provides much less separation. Percentages use all 203 training rows per class; respondents could select multiple seasons.



Figure 1. Season selection rates by sport in the training split.

Text also shows class-specific evidence: "blue", "orange", and "green" have the highest mean colour TF-IDF weights for Baseball, Basketball, and Soccer, respectively. Soccer song responses prominently contain "waka" and "shakira"; Baseball responses contain words from "Take Me Out to the Ball Game". "Bored" ranks highest for Baseball viewing feelings, whereas "excited" ranks highest for Basketball and Soccer. These are descriptive associations, not validated feature-importance estimates.

The combined representation contains 390 features for 609 training rows. This motivates regularization and controlled model complexity. Distributed word evidence supports trying a regularized linear classifier, while option-specific and potentially nonlinear relationships support tree-based alternatives. Their predictive value will be assessed using held-out validation data.